“Interior-Point Method for Solving Linearly Constrained Convex Optimization Problems”

A

Project Report

submitted in partial fulfilment of the

requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING-AIML/AIDE

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CANDIDATE'S DECLARATION

We hereby certify that the project work entitled **" Interior-Point Method for Solving Linearly Constrained Convex Optimization Problems"** in partial fulfilment of the requirements for the award of the Degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING with specialization in AIML and submitted to the Department of Computer Science & Engineering at Faculty of Engineering and Technology, Jain (Deemed to be University) Bengaluru, is an authentic record of our work carried out during a period from **Feb 2026 to March 2026** under the supervision of **Dr. Shivam Swarup, Assistant Professor.**

The matter presented in this project has not been submitted by us for the award of any other degree of this or any other University.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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ACKNOWLEDGEMENT

We wish to express our deep gratitude to our guide **Dr. Shivam Swarup**, for all advice, encouragement and constant support he has given us throughout our project work. This work would not have been possible without his support and valuable suggestions.

We sincerely thank to our respected Program Head Dr. Chandrasekar V for his great support in doing our project in **Interior-Point Method for Solving Linearly Constrained Convex Optimization Problems**.

We would like to thank all our **friends** for their help and constructive criticism during our project work. Finally, we have no words to express our sincere gratitude to our **parents** who have shown us this world and for every support they have given us.

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ABSTRACT

Optimization is an important aspect of machine learning, artificial intelligence, and many engineering fields. Many optimization tasks require meeting certain constraints while minimizing or maximizing an objective function. However, standard optimization techniques like Gradient Descent don't naturally handle constraints, which can sometimes result in solutions that don't satisfy those constraints. In this project, we address a convex optimization problem with linear constraints by applying three different methods: Gradient Descent, Projected Gradient Descent, and the Interior-Point Method. The objective function here is quadratic and convex, and the constraints define a feasible region limited by a linear boundary. The Interior-Point Method uses a barrier function to keep the solution from crossing constraints as it searches for the optimum. The algorithms were implemented in Python, and we used graphs to illustrate how each method converges and traces the optimization path. The results show that the Interior-Point Method consistently respects the constraints throughout the process and reaches the optimum in a steady manner. Overall, this work demonstrates that interior-point methods are effective for solving constrained optimization problems and offers insight into their strengths compared to traditional gradient-based techniques. The project also highlights the significance of optimization methods that take constraints into account in current computational challenges. By comparing various optimization techniques, the study shows that the Interior-Point Method effectively integrates constraints into the optimization using barrier functions. Graphical analyses of convergence and optimization paths offer a clearer understanding of how each algorithm behaves throughout the optimization. The results indicate that interior-point methods are both reliable and efficient for handling constrained convex optimization problems, and they have potential applications in more advanced areas like machine learning, operations research, and artificial intelligence.

Keyword: Convex Optimization, Interior Point Method, Gradient Descent, Constrained Optimization, Barrier Functions

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1. INTRODUCTION

1.1. An Introduction

Optimization plays a key role in mathematics, machine learning, and engineering. It involves finding the best possible solution to a problem by minimizing or maximizing an objective function while meeting certain constraints. Many practical problems—like allocating resources, training machine learning models, or optimizing financial portfolios—require solving optimization problems with constraints. Convex optimization refers to a particular group of problems where both the objective function and the feasible set are convex. These problems are significant because they often allow us to find a global optimum. Yet, when constraints are involved, standard methods like Gradient Descent may produce solutions that don't satisfy those constraints.

To handle this, more advanced techniques such as Projected Gradient Descent and Interior-Point Methods are employed. Projected Gradient Descent maintains feasibility by projecting any solution outside the feasible region back within it. Meanwhile, the Interior-Point Method uses barrier functions to keep the solution away from the edges of the permissible region. In this project, we explore and compare three optimization methods designed to solve a convex problem with linear constraints. The algorithms are implemented in Python, and we study their convergence patterns and optimization trajectories through various visualizations.

1.2. Project Idea

This project focuses on comparing various optimization methods for solving constrained convex problems. The chosen objective function is a simple quadratic form, which makes it easier to visualize and analyze how the optimization proceeds.

Three optimization algorithms are implemented:

1. Gradient Descent
2. Projected Gradient Descent
3. Interior-Point Method

All the algorithms are tested on the same problem, and their performance is evaluated through convergence plots and visual representations of their optimization paths. The results highlight that the Interior-Point Method is effective at keeping the solution within feasible bounds and achieves steady convergence when tackling constrained

1.3. MOTIVATION

In many real-world optimization problems, there are strict constraints that must be met. Traditional optimization algorithms often focus on problems without constraints and may struggle to enforce these limits effectively. Interior-point methods offer a practical mathematical strategy for handling constrained optimization by using barrier functions that keep the solution within the allowed boundaries. These methods have found broad application in areas like linear programming, nonlinear optimization, and machine learning.

This project aims to explore how interior-point methods operate and to evaluate their performance compared to other commonly used optimization algorithms. One of the reasons for this project is to gain hands-on experience with how theoretical optimization ideas work in actual computational problems. Although many optimization methods are explored through math, coding them offers a clearer view of how they behave, how they converge, and what their limits are. By looking at the optimization paths and convergence charts, it becomes easier to see how different algorithms move toward the best solution and manage constraints during the process. This practical work helps connect the theory of numerical optimization with how it's applied in real cases like machine learning and engineering.

1.4. SCOPE

The scope of this project includes:

- Understanding the concept of convex optimization and constrained optimization.
- Studying the mathematical formulation of optimization problems with linear constraints.
- Implementing optimization algorithms using Python and numerical computation libraries.
- Comparing Gradient Descent, Projected Gradient Descent, and Interior-Point Method.
- Understanding how barrier functions help maintain feasibility in constrained optimization problems.
- Visualizing convergence behavior using graphs.
- Analyzing the performance of different optimization techniques.

This project focuses on a simple convex optimization problem to clearly demonstrate the behavior of different optimization algorithms and how constraints influence the optimization process.

2. RELATED WORK

Optimization methods have been extensively studied in mathematics, operations research, and machine learning. These methods are essential for solving problems that require minimizing or maximizing an objective function while meeting certain constraints. Over time, many techniques have been developed to tackle different types of optimization problems, such as unconstrained optimization, constrained optimization, linear programming, and nonlinear programming. Gradient-based algorithms like Gradient Descent are widely used for unconstrained optimization because they are simple and computationally efficient. Gradient Descent works by iteratively adjusting the solution in the direction opposite to the gradient of the objective function.

Projected Gradient Descent builds on standard Gradient Descent to handle constraints. After each gradient step, if the solution violates any constraints, it is projected back into the feasible region. This process ensures the optimization stays within valid limits. Projected Gradient Descent is commonly applied in constrained machine learning problems, signal processing, and convex optimization where maintaining feasible solutions is critical. Interior-point methods were developed as an efficient alternative to traditional simplex methods in linear programming. Unlike approaches focusing on boundaries, interior-point methods work inside the feasible region and move gradually toward the optimum.
These interior-point algorithms have been effectively used in various areas including linear, quadratic, and nonlinear programming, as well as large-scale optimization. Their ability to efficiently handle complex constraints has made them popular in engineering design, machine learning model tuning, financial portfolio optimization, and resource allocation. Researchers like Boyd and Vandenberghe have made significant contributions to convex optimization and interior-point methods, providing both theoretical foundations and practical algorithms for constrained problems.

This project examines and compares these optimization methods to better understand how they perform on a linearly constrained convex optimization problem. By studying Gradient Descent, Projected Gradient Descent, and Interior-Point methods, the project sheds light on how different strategies manage constraints and approach the optimal solution.

3. PROBLEM STATEMENT/OBJECTIVES

In many optimization tasks, the aim is to minimize an objective function while meeting specific constraints. These types of problems often arise in fields like machine learning, engineering design, and operations research. Effectively managing constraints during optimization can be challenging since many standard algorithms are mainly built for problems without constraints. This project focuses on a convex optimization problem with linear constraints. Specifically, the goal is to minimize a quadratic convex function while making sure the solution adheres to a linear inequality constraint.

The optimization problem is mathematically defined as:

Minimize:

$$f(x) = x_1^2 + x_2^2$$

Subject to the constraint:

$$x_1 + x_2 <= 1$$

where

$x = (x_1, x_2)$represents the decision variables of the optimization problem.

The objective function $f(x)$is a convex quadratic function, which guarantees that the global minimum can be found within the feasible region defined by the constraint.

The constraint $x_1 + x_2 \leq 1$restricts the solution space to a specific feasible region.

The main objectives of this project are:

- To implement Gradient Descent for solving the optimization problem.
- To implement Projected Gradient Descent to ensure that the solution satisfies the constraint by projecting infeasible solutions back into the feasible region.
- To implement the Interior-Point Method using barrier functions to maintain feasibility during the optimization process.
- To compare the convergence behavior of the three optimization algorithms.
- To visualize the optimization paths and convergence graphs in order to analyze the performance of the algorithms.

Through this study, the project aims to understand how different optimization techniques handle constraints and how efficiently they converge toward the optimal solution.

4. IMPLEMENTATION AND RESULTS

4.1. IMPLEMENTATION CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Objective Function
def objective(x):
    return x[0]**2 + x[1]**2

def gradient(x):
    return np.array([2*x[0], 2*x[1]])

# Constraint: x1 + x2 <= 1
def constraint(x):
    return 1 - (x[0] + x[1])

# Optimizer 1: Gradient Descent
def gradient_descent(x0, lr=0.1, iters=50):

    x = x0.copy()
    history = []

    for i in range(iters):

        grad = gradient(x)
        x = x - lr * grad

        history.append(x.copy())

    return x, np.array(history)

# Optimizer 2: Projected Gradient
def projected_gradient(x0, lr=0.1, iters=50):
```

```
    x = x0.copy()
    history = []

    for i in range(iters):

        grad = gradient(x)
        x = x - lr * grad

        # projection if constraint violated
        if x[0] + x[1] > 1:
            x = x / (x[0] + x[1])

        history.append(x.copy())

    return x, np.array(history)

# Optimizer 3: Interior Point Method
def interior_point(x0, mu=0.1, lr=0.05, iters=50):

    x = x0.copy()
    history = []

    for i in range(iters):

        # ensure constraint value doesn't become zero
        c = max(constraint(x), 1e-6)

        grad_f = gradient(x)

        # barrier gradient
        barrier_grad = mu * np.array([1/c, 1/c])

        grad = grad_f + barrier_grad
```

```
        x = x - lr * grad

        history.append(x.copy())

    return x, np.array(history)

# Run Experiments
x0 = np.array([0.5, 0.3])

gd_sol, gd_hist = gradient_descent(x0)
pgd_sol, pgd_hist = projected_gradient(x0)
ip_sol, ip_hist = interior_point(x0)

print("Gradient Descent Solution:", gd_sol)
print("Projected Gradient Solution:", pgd_sol)
print("Interior Point Solution:", ip_sol)

# Compute Loss Values
def compute_loss(history):

    loss = []

    for x in history:
        loss.append(objective(x))

    return loss

loss_gd = compute_loss(gd_hist)
loss_pgd = compute_loss(pgd_hist)
loss_ip = compute_loss(ip_hist)

# Plot Convergence Graph
plt.figure(figsize=(8,5))
```

```
plt.plot(loss_gd, label="Gradient Descent")
plt.plot(loss_pgd, label="Projected Gradient")
plt.plot(loss_ip, label="Interior Point")

plt.xlabel("Iterations")
plt.ylabel("Objective Value")
plt.title("Optimizer Convergence Comparison")

plt.legend()
plt.grid()
plt.show()

# Plot Optimization Paths
plt.figure(figsize=(6,6))

gd_hist = np.array(gd_hist)
pgd_hist = np.array(pgd_hist)
ip_hist = np.array(ip_hist)
plt.plot(gd_hist[:,0], gd_hist[:,1], 'r-o', label="Gradient Descent")
plt.plot(pgd_hist[:,0], pgd_hist[:,1], 'g-s', label="Projected Gradient")
plt.plot(ip_hist[:,0], ip_hist[:,1], 'b-^', label="Interior Point")

# constraint boundary
x = np.linspace(0,1,100)
y = 1 - x
plt.plot(x, y, 'k--', label="Constraint Boundary")
plt.xlabel("x1")
plt.ylabel("x2")
plt.title("Optimization Paths")
plt.legend()
plt.grid()

plt.show()
```

4.2. RESULTS:

OUTPUT:

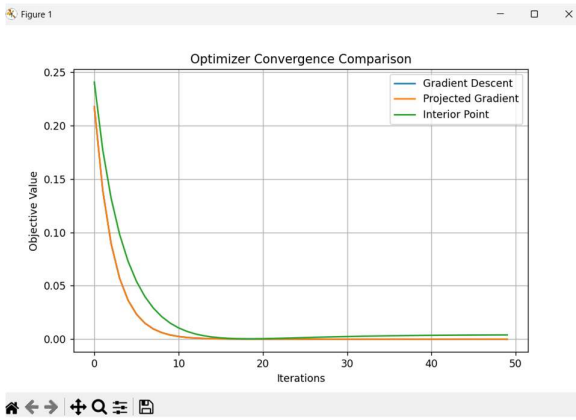

Fig 1: Optimizer Convergence Comparison

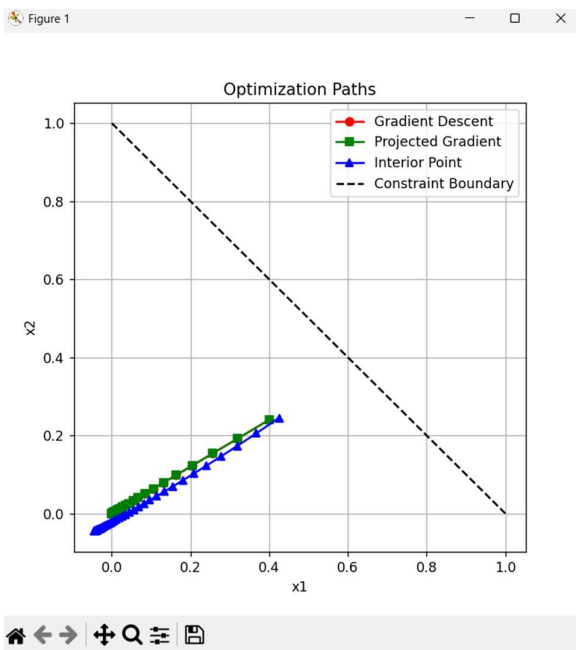

Fig2: Optimization Paths

```
C:\Users\kuruv\AppData\Local\Programs\Python\Python312\python.exe
Gradient Descent Solution: [7.13623846e-06 4.28174308e-06]
Projected Gradient Solution: [7.13623846e-06 4.28174308e-06]
Interior Point Solution: [-0.04404478 -0.04507554]

Process finished with exit code 0
```

Fig 3: Optimizer Solution Result Comparison

5. CONCLUSION

This project examined several optimization algorithms for solving convex problems with linear constraints. We implemented three approaches: Gradient Descent, Projected Gradient Descent, and the Interior-Point Method. The results show that Gradient Descent converges quickly but does not maintain the constraints. Projected Gradient Descent keeps the solution feasible by projecting it back into the allowed region, though this can cause some fluctuations near the boundaries.

The Interior-Point Method uses a barrier function to prevent the solution from crossing constraint limits. This helps the optimization stay within the feasible region and converge smoothly toward the best solution. Comparing the optimization paths and convergence patterns indicates that the Interior-Point Method offers more stable and effective performance for constrained problems.

In summary, the project underscores the importance of properly handling constraints in optimization and demonstrates that interior-point methods can solve constrained convex problems efficiently.

6. REFERENCES

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